Carlo Felipe Vivanco Coronado

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PROFILE

Mechatronics and AI engineer with an MSc in Intelligent Systems & Robotics. Trained **deep RL** and **imitation learning** policies for a Franka Panda in Isaac Sim, and deployed them **sim-to-real** through ROS 2. Brings 3+ years of production, safety-critical test software in **Python, C++ and Go** across Tesla, Ford and Stellantis. Now builds LLM, multi-agent and MLOps systems on GCP at Nexxia, including a model router that **cut LLM cost per request 96%**.

EDUCATION

University of Essex October 2024 – November 2025
Master of Science degree with Distinction in Intelligent Systems and Robotics
Coursework: Intelligent Systems & Robotics (Fuzzy Logic), Deep Learning & Neural Networks, Machine Learning, Computer Vision, Electronic Design, Embedded Systems with RTOS, Microcontrollers (UART, SPI, I2C, Ethernet).

Tecnológico de Monterrey August 2021 – June 2026
Bachelor of Science in Mechatronics Engineering, GPA: 95/100 *Graduated with Honors*
Coursework: Mechanical & Electronic Design (CAD, PCBA), Control Systems, Industrial Automation, Robotics, Analog & Digital Systems, CNC & PLC Programming, Project Management.

EXPERIENCE

Nexxia February 2026 – Present
AI Software Engineer | Full-Time, On-Site *Mexico City, Mexico*

- Architect and lead full-stack development on **GCP**: **FastAPI** and **React/TypeScript** on Cloud Run, a **Zero Trust** security model (Cloud IAP + JWT, multi-tier RBAC) and GitLab CI with security gates.
- Designed an **embedding-KNN model router** tuned with **Optuna** across **7 models**, **cutting LLM cost per request 96%** vs. an always-frontier baseline, with equal or better quality on production queries.
- Build a **multi-agent architecture** (**Vertex AI**, **Google ADK**) with **RAG** pipelines and an **MCP** tool layer.

Tesla August 2025 – December 2025
Software Engineering Manufacturing Test Intern | Full-Time, On-Site *Palo Alto, California, USA*

- Shipped automated ECU code-generation tooling and led **root cause analysis** on integration failures, **cutting firmware test development time 25% per release**.
- Built test frameworks in **C++**, **Python** and **Go**, automating high-voltage battery validation over **CAN/UDS** on **hardware-in-the-loop (HIL)** rigs.
- Ran schematic analysis, **board bring-up** and **calibration** alongside embedded teams, delivering modular HV battery test platforms through **end-of-line (EOL)** validation.

Ford Motor Company June 2024 – September 2024
Software Integration Automation Trainee | Part-Time, Hybrid *Mexico City*

- Built Python, TestStand, LabVIEW and PyTest frameworks with GitHub CI/CD **regression testing**, automating vehicle-controls validation across **SIL, MIL and HIL**.
- Integrated test pipelines with break-out box (BOB) setups, validating system-level vehicle controls over **CAN, LIN, SOME/IP and UDS**.
- Drove **root cause analysis** on integration defects, improving traceability and reliability across Design Verification stages.

Ford Motor Company June 2023 – June 2024
Feature Systems Engineering Trainee | Part-Time, Hybrid *Mexico City*

- Built **MATLAB/Simulink digital twins** with state-machine controls (e.g., Trailer Brake Control), locking vehicle-feature control logic before hardware integration.
- Built breadboard and on-vehicle test automation, **cutting system validation time 50%**. Earned the High Achiever Badge in Ford's Intern City of Tomorrow Challenge.

Stellantis January 2023 – June 2023
Electrical Quality Engineering Intern | Part-Time, Remote *Mexico City*

- Implemented quality-system improvements (JIRA, Power BI, Excel) on production lines, **cutting issue processing time 30%** and surfacing high-impact GIMs earlier.

PROJECTS

Deep RL & Imitation Learning for Robotic Manipulation (Sim-to-Real) November 2024 – July 2025
Master's Dissertation, University of Essex | PyTorch, Stable-Baselines3, Isaac Sim / Isaac Lab, ROS 2

- Built an **Isaac Sim / Isaac Lab** Franka Panda bolt-removal task with a **shaped multi-stage reward** and benchmarked **5 RL algorithms**, identifying TD3 as the highest-reward and most stable.
- Applied **Optuna**, **curriculum learning** and **behavior-cloning** pre-training with RL fine-tuning, breaking a multi-week plateau and converging TD3 over **15M steps on 8 parallel CUDA environments**.
- Built a custom **ROS 2** node fed with 3D bolt poses from fine-tuned **YOLOv8-seg** and **RealSense RGB-D**, transferring the policy **sim-to-real** to a physical **Franka Panda**.

Human Intention Recognition for Human-Robot Interaction October 2024 – March 2025
MSc Team Project (7 engineers), University of Essex | ROS, Gazebo, YOLO, MediaPipe, Transformers

- Integrated perception, speech and robot control in **ROS/Gazebo**, delivering a **Franka Panda** assistant that passes requested tools into a worker's open hand and takes them back.
- Fused **YOLO**, **MediaPipe** hand landmarks and zero-shot **NLP intent classification** (BART-MNLI), reaching **>95% gesture**, **>90% tool** and **95% speech-command** accuracy.
- Trained a **reinforcement learning** policy for gripper alignment on RGB-D camera-to-robot transforms, positioning the gripper within **<5 mm** of target in simulation.

E-Racing CEM, Body & Exteriors Design Leader August 2021 – Present
Tecnológico de Monterrey | Shell Eco-Marathon Americas Prototype Electric

- Led design-to-manufacture of the carbon-fiber aerodynamic body in **SolidWorks** and **CATIA**, hitting competition drag and downforce targets.
- Designed and validated motor-controller **PCBAs** with the electronics team using **DFM** principles, reducing failures in competition runs.

TECHNICAL SKILLS

Programming: Python (NumPy, PyTorch, PyTest, FastAPI), modern C++, Go, CUDA, MATLAB, SQL, Linux, Git
Robot Learning: Deep RL (TD3, SAC, PPO, DDPG), Imitation Learning / Behavior Cloning, Sim-to-Real, Reward Shaping, Curriculum Learning, Parallel-Env Training, Optuna, Stable-Baselines3
Perception & 3D Vision: YOLOv8 (detection/segmentation), MediaPipe, OpenCV, Intel RealSense RGB-D, camera-to-robot transforms
Simulation & Robotics: NVIDIA Isaac Sim / Isaac Lab, MuJoCo, Robosuite, Gazebo, ROS / ROS 2, Franka FCI
NLP / LLMs: Hugging Face Transformers, zero-shot classification, LLMs, RAG, multi-agent systems (LangGraph, Google ADK)
Embedded & Hardware: HIL, SIL/MIL, EOL testing, CAN/LIN/SCPI/UDS, EtherCAT, RTOS
Cloud / MLOps: GCP (Vertex AI Pipelines, Cloud Run), Docker, Kubernetes, CI/CD (GitHub Actions, Terraform)
Languages: Spanish (Native), English (C1, IELTS Advanced), German (A2)